

1. In your perspective, what could possibly happen if the number of customers doubled and the number of servers remains the same?

If the number of customers doubles while the number of servers stays the same, the queue length will increase a lot. This means more people will be waiting in line, so the queue time will also become longer. The servers will become busier, and the service time per customer stays the same, but overall waiting becomes worse. This can cause delays and may even lead to customer dissatisfaction. In extreme cases, the system can become overloaded and inefficient.

2. Where would you categorize the arrival process of the given queuing system, under the infinite population model or under the finite population model?

The system is best described as an infinite population model. This is because customers can come randomly and are not limited to a fixed number of people. In real-life situations like stores or service centers, new customers can always arrive at any time. There is no fixed group being served repeatedly. That is why it fits the infinite population model.

3. Will you agree that the system above encompasses a specific queue discipline?

Yes, the system uses a queue discipline, most likely First Come, First Served. This means customers are served in the order they arrive. The diagram shows people lining up and getting served one by one, which follows this rule. Queue discipline is important because it ensures fairness and order in the system.

4. What would you suggest to improve the queuing system above?

One way to improve the system is to add more servers, so more customers can be served at the same time. Another way is to reduce service time, maybe by making processes faster or more efficient. The system can also use a priority queue for urgent cases. Adding a better scheduling system or using technology can help reduce crowding. These improvements will reduce queue time and make the system faster and more efficient.